

Saic Institute of Management & Technology (SIMT)
University of Dhaka (Affiliated Engineering College)
Department of Computer Science and Engineering

1st Year 1st Semester

Course Code: CHE-1104

Course Name: Chemistry

Marks: Total 25 Time: 50 minute

1. Describe the postulates of Bohr atomic model. Mention its limitations 5
2. What is quantum number? Why are four quantum numbers required to explain the position of an electron in an atom? 1+5
3. Discuss the following rules for electronic configuration of atom with example: 6
 - i) Pauli's exclusion principle
 - ii) Hund's rule
4. Explain Heisenberg's Uncertainty Principle. Show that the principle is valid only for small particles and not for large particles. 4
5. Write down the electronic configuration of - 2
 - Fe^{2+} , Cr , S^{2-} and Br^-
6. Explain why the last electron of K enters into the 4s orbital but not in the 3d orbital? 2

(19) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d$